

VALID: A 12-Item Validation Checklist for Financial Machine Learning

Jaewook Kim

Independent Researcher

Goyang-si, Republic of Korea

jwim1101@gmail.com

Abstract

Machine learning strategies for cryptocurrency trading consistently report exceptional backtest results, yet practitioners consistently fail to replicate them. No domain-specific validation standard exists for financial ML. We introduce VALID, a 12-item validation checklist—to our knowledge the first checklist-style protocol for financial ML to pair binary statistical and economic gates in a fixed reporting order, addressing temporal leakage, backtest overfitting, cost modeling, and class imbalance.

An audit of 80 published papers (74 empirical) reveals the median study satisfies only 2.5 of 12 items. Applying VALID to 340 strategy variants across three assets and four timeframes, we document three failure modes: directional prediction bias (90–97%), statistical-economic disconnect, and cost illusion (net Sharpe falling monotonically with trading frequency). Monte Carlo analysis ($n = 200$) shows 27% false positive rates under standard validation, reduced to 0% by CPCV with PBO. Even strategies surviving multiple testing correction exhibit $PBO = 1.0$ (complete overfitting) under CPCV. We outline why extending validation to LLM-based trading agents—whose non-deterministic generation resists the checklist’s programmatic criteria—remains an open problem.

CCS Concepts

• **Mathematics of computing** → Probabilistic algorithms; • **Computing methodologies** → *Neural networks*.

Keywords

financial machine learning, validation framework, backtest overfitting, cryptocurrency trading, reporting standard

1 Introduction

Our audit of 80 published cryptocurrency ML papers (74 empirical) reveals that the median study satisfies only 2.5 of 12 basic validation items across the empirical subset. The consequences are predictable: strategies reporting exceptional in-sample performance consistently fail out-of-sample, replicating the broader pattern identified by Ioannidis [14] in biomedical research and by Harvey et al. [13] in equity factor literature, where the majority of 316 published factors are likely false discoveries. McLean and Pontiff [20] documented 58% post-publication decay in return predictability; Wiecki

et al. [24] analyzed 888 algorithmic strategies and found near-zero correlation between backtest and live Sharpe ratios.¹

Domain-specific reporting checklists have demonstrably improved research quality in other fields. TRIPOD [8] (22 items, 10,000+ citations) transformed clinical prediction methodology. REFORMS [15] addresses general ML science with 32 items. Yet **no published standard addresses financial ML**—a domain with unique challenges including temporal data dependencies, transaction cost modeling, structural class imbalance, and the distinction between statistical and economic significance.

This paper makes three contributions:

- (1) **VALID**: a 12-item validation checklist for financial ML—to our knowledge the first checklist-style protocol pairing binary statistical and economic gates for this field, complementing rather than superseding prior backtesting guidance (Figure 1).
- (2) **Three failure modes at scale**: using 340 strategy variants across three assets and four timeframes, we document directional prediction bias, statistical-economic disconnect, and cost illusion.
- (3) **Monte Carlo FPR quantification**: AUC-based validation produces 27% false positives; CPCV with PBO reduces this to 0%. Multiple testing correction (Bonferroni) leaves nine “significant” strategies—all with $PBO = 1.0$, demonstrating that standard corrections alone cannot detect backtest overfitting.

2 Related Work

2.1 Backtest Overfitting and Validation Methodology

The theoretical foundations for detecting backtest overfitting are well established. The Reality Check of White [23] and the Superior Predictive Ability test of Hansen [12] address data snooping in multiple hypothesis testing. Bailey et al. [4] introduced the Probability of Backtest Overfitting (PBO) and the Deflated Sharpe Ratio (DSR), providing the first formal tools for quantifying strategy selection bias. Arian et al. [2], published in *Knowledge-Based Systems*, compared CPCV against walk-forward methods in a synthetic controlled environment, confirming CPCV’s superior false-positive control. Witzany [25] provided the only formal critique of PBO, demonstrating that it exhibits negative bias when strategy

¹An extended version with full experimental detail is available as SSRN Working Paper No. 6508779.

returns are similar; Section 4.4 reports empirical evidence consistent with that analysis.

Empirically, Wiecki et al. [24] analyzed 888 algorithmic trading strategies from the Quantopian platform and found that backtest Sharpe ratio has near-zero predictive power for out-of-sample performance ($R^2 < 0.025$). Arnott et al. [3] proposed a backtesting protocol with multiple testing safeguards, though without a domain-specific checklist. Our analysis of 340 ML cryptocurrency strategies follows an analogous large-cohort evaluation design.

2.2 Reporting Standards for ML Research

Domain-specific reporting checklists have demonstrated substantial impact on research quality. In clinical prediction, TRIPOD [8] (22 items) has accumulated over 10,000 citations and measurably improved methodological transparency. Its ML extension, TRIPOD+AI [7] (27 items), and the LLM-specific TRIPOD-LLM [9] (19 items, 50 subitems) further broaden coverage. In general ML research, REFORMS [15] proposes 32 items, while Model Cards [21] and Datasheets for Datasets [10] address model and data documentation respectively—each accumulating thousands of citations without proposing novel algorithms.

Critically, **no published reporting standard addresses financial ML**. Neither REFORMS nor TRIPOD+AI covers transaction cost modeling, class imbalance in directional prediction, temporal cross-validation with purge/embargo, or the distinction between statistical and economic significance. VALID fills this gap.

2.3 The “Most Research Is Wrong” Tradition in Finance

VALID enters a well-established intellectual lineage. Harvey et al. [13] cataloged 316 published equity factors and argued that most are likely false discoveries, requiring t -statistics above 3.0. McLean and Pontiff [20] documented 58% post-publication decay in return predictability. Novy-Marx and Velikov [22] showed that most equity anomalies with high turnover become unprofitable after costs—a finding we extend to cryptocurrency ML in Section 4.3. More recently, Karl et al. [17] argued at ICML 2024 that negative results in ML prevent community-wide resource waste, providing direct methodological support for VALID’s approach. Successful negative-result papers in adjacent domains—such as Grinsztajn et al. [11] (trees outperform deep learning on tabular data) and Zeng et al. [26] (linear models outperform Transformers for time series)—demonstrate that rigorous disconfirmation, paired with constructive alternatives, consistently achieves high impact.

In cryptocurrency specifically, the validation landscape remains barren. Our audit of 80 papers (Section 4.1) finds 0% CPCV adoption, 73% class imbalance neglect, and 54% cost omission. VALID addresses this gap by synthesizing the theoretical tools of Bailey and López de Prado, the large-cohort evaluation methodology of Wiecki et al., and the “most findings are false” framing of Harvey et al.—applied to

cryptocurrency ML trading, a field where our audit reveals systematic validation gaps.

3 The VALID Framework

3.1 Design Principles

VALID is designed to be finance-specific, evidence-based, and actionable. Each item addresses a failure mode quantified in Section 4. Items are binary (pass/fail) to enable objective assessment and cross-study comparison.

Item selection follows two criteria. First, each item corresponds to a deficiency observed at high frequency in the 80-paper audit (V7–V8 target the 54% cost-omission rate; V2 targets the 73% class-imbalance rate). Second, each item has an established methodological basis in prior protocols [3], including combinatorial purged cross-validation and deflated performance metrics. The checklist has not yet undergone formal review by independent practitioners; we regard this version as a v1.0 proposal and invite community revision, following the precedent of TRIPOD’s iterative development.

3.2 The 12 VALID Items

Table 1 presents the complete checklist. We describe each item below.

V1–V2 (Class Distribution and Balancing). Cryptocurrency markets trend upward over most historical windows, producing structurally imbalanced labels (e.g., 60–70% “long” in Triple Barrier labeling). Without explicit reporting (V1) and balancing (V2), models learn the base rate rather than genuine directional signals. Section 4.2 quantifies this: unbalanced CatBoost predicts “long” 90–97% of the time.

V3 (Temporal Splitting). Random train-test splits create temporal leakage: future information contaminates training data through autocorrelated features and overlapping lookback windows. Temporal splitting with purge and embargo gaps [19] eliminates this. Our audit finds 7% of papers still use random splits.

V4 (CPCV with PBO). Single train-test splits or k -fold cross-validation provide one or k performance estimates, insufficient for detecting overfitting across large strategy search spaces. CPCV with N groups and k test groups generates $\binom{N}{k}$ unique combinations and $\varphi = (k/N)\binom{N}{k}$ non-overlapping paths; PBO measures the fraction where in-sample winners underperform out-of-sample. Section 4.5 shows that removing V4 raises FPR from 0% to 27%—the single most impactful item.

V5 (Parameter-Space Variance). $PBO = 0$ can arise when all configurations perform similarly and the search cannot distinguish between them, rather than from a genuinely robust signal. Reporting $\text{Var}(\text{SR}_{\text{IS}})$ and comparing it against a Monte Carlo null distribution distinguishes these cases (Section 4.4).

V6 (Permutation Tests). A strategy may appear profitable on structured data yet fail to exceed performance on randomly permuted labels. At least 100 permutations are needed to estimate the null distribution with reasonable precision.

V7–V8 (Cost Modeling and Sensitivity). Transaction costs are the single largest source of alpha erosion in ML trading strategies. V7 requires reporting net-of-cost performance; V8 requires demonstrating that conclusions are robust across plausible cost assumptions (e.g., 5–50 bp). Section 4.3 shows net Sharpe falling monotonically with trading frequency, and 82% gross-to-net Sharpe erosion in the one run where both are measured at 18 bp.

V9 (Baseline Comparison). 35% of audited papers use only buy-and-hold as a baseline—a passive strategy that any trend-following rule can beat in trending markets. V9 requires comparison against simple active strategies (e.g., SMA crossover, RSI, momentum) to establish genuine ML value-added.

V10 (Regime Evaluation). Strategies trained on bull-market data may fail during bear markets or regime transitions. V10 requires explicit evaluation across market regimes. Our regime-conditional analysis (Section 4.3) confirms the benchmark outperforms buy-and-hold in both calm and stressed regimes.

V11–V12 (Trade Frequency and Code Availability). V11 requires reporting trade frequency to assess cost headwinds. V12 requires code availability: 85% of audited papers provide no code, making reproduction impossible.

Items V1–V6 constitute *Stage 1 (Statistical)*: validation requirements verifiable from the manuscript alone. Items V7–V12 form *Stage 2 (Economic)*: validation requirements for strategies intended for live trading. A study passing Stage 1 but failing Stage 2 exemplifies the *statistical-economic disconnect*—statistically significant signals with no exploitable alpha.

3.3 Comparison with Existing Standards

Table 2 positions VALID relative to established checklists. VALID supplements REFORMS with financial-specific items (cost modeling, class balance, CPCV) absent from general-purpose frameworks.

4 Empirical Validation

We evaluate VALID through five complementary analyses: a literature audit (Section 4.1), bull bias (Section 4.2), cost illusion with regime conditioning (Section 4.3), statistical-economic disconnect (Section 4.4), and ablation with Monte Carlo and multiple testing (Section 4.5).

Experimental Setup. The experimental pipeline spans 340 strategy variants² constructed from: three cryptocurrency assets (BTC, ETH, SOL); four timeframes (15-minute, 1-hour, 4-hour, daily); and eight model families (CatBoost, LightGBM, Random Forest, Transformer, LSTM, TCN, 2-layer MLP, 3-layer MLP). Eleven component types are systematically varied: SHAP-based feature selection rules, order flow

²The corpus includes 18 cost-level entries (a rule-based benchmark and two signal proxies evaluated at six cost levels each). Excluding them, 55.0% of the remaining 322 variants are negative net of costs and 11 exceed the benchmark, one of which is the benchmark’s own entry in the baseline block; all corpus-level conclusions are unchanged.

features, external macro features (DXY, gold, NASDAQ), the complete ML pipeline, parameter sensitivity analysis, cross-section momentum, cost sensitivity, deep learning architectures, VALID ablation studies, volatility overlays, and bull-bias analysis (Table 11, rows (a)–(k)).

All variants use a common infrastructure: CUSUM-triggered Triple Barrier labeling [19] (Appendix E) for event-driven sampling, expanding-window training with purged and embargoed temporal splitting (V3), and CPCV with $N = 6$ chronologically ordered groups and $k = 2$ test groups, generating $\binom{6}{2} = 15$ unique train-test combinations and $\varphi = 5$ non-overlapping backtest paths for PBO computation (V4). Transaction costs are fixed at 18 bp round-trip (conservative for Binance VIP-0 taker fees on hourly and daily frequencies). The benchmark is a Dual Momentum (DM) 252/126 [1] strategy with a drawdown-based risk overlay, achieving $SR = 0.917$ after costs—a deliberately strong baseline (V9) that exceeds buy-and-hold ($SR = 0.618$), SMA 200 crossover (0.723), and RSI(14) (0.804).

4.1 Literature Audit

We systematically audit 80 cryptocurrency ML trading papers published between 2018 and 2026, sourced from Google Scholar, Scopus, and Web of Science. Of the 80 papers, 74 are empirical; six are excluded (two surveys, two synthetic-only methodological studies, one equities-only study, one non-ML microstructure study). We classify a paper as empirical if it applies a machine learning method to real cryptocurrency market data for a trading or prediction task. We code the 74 empirical papers across seven dimensions mapped to VALID items. Table 3 reports the results.

The median study satisfies only 2.5 of 12 VALID items. The most striking finding is that *zero* studies employ CPCV (V4)—the single validation component that our ablation study identifies as most impactful (Section 4.5). 73% neglect class imbalance, a failure that produces the bull bias documented in Section 4.2. 54% omit transaction costs entirely, enabling the cost illusion of Section 4.3. 85% provide no code, making independent verification impossible.

These findings are consistent with the broader reproducibility concerns raised by Kapoor and Narayanan [16], who identified data leakage affecting 294 studies across 17 scientific fields. Cryptocurrency ML appears particularly susceptible: the combination of trending markets (structural class imbalance), 24/7 trading (high-frequency cost headwinds), and rapid publication cycles creates ideal conditions for the failure modes VALID is designed to detect.

4.2 Bull Bias

Without class balancing, tree-based models learn the base rate of trending cryptocurrency markets, predicting “long” 90–97% of the time. Table 4 demonstrates this across three assets.

Table 1: The 12 VALID items. Each item addresses a specific failure mode quantified in Section 4.

#	Item	Rationale	Failure Mode
V1	Report prediction class distribution	Base-rate learning on trending assets	Bull bias
V2	Test with/without class balancing	Structural class imbalance in crypto	Bull bias
V3	Use temporal splitting only	Information leakage across time, including feature lookback windows	Temporal leakage
V4	Apply CPCV with PBO	Holdout insufficient for large search	Overfitting
V5	Report $\text{Var}(\text{SR}_{15})$	$\text{PBO} = 0$ from flat landscape (Witzany, 2021)	PBO misinterpretation
V6	Include permutation tests (≥ 100)	Confirm signal existence	Spurious patterns
V7	Report net performance with costs	Cost omission inflates alpha	Cost illusion
V8	Cost sensitivity analysis	Cost assumptions can flip alpha sign	Cost illusion
V9	Compare against simple baselines	Weak baselines inflate relative performance	Weak baselines
V10	Evaluate in bear markets and regime transitions	Bull-market trend-following not informative	Regime overfit
V11	Report trade frequency	High-frequency cost headwinds	Hidden turnover
V12	Provide code for reproducibility	$<10\%$ of papers provide code	Irreproducibility

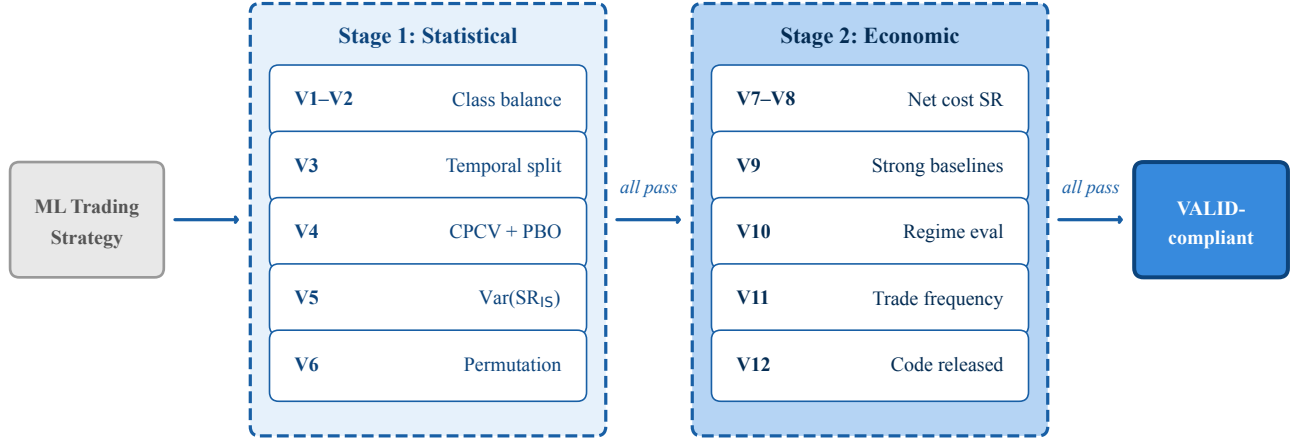


Figure 1: The VALID framework. A candidate financial ML strategy is screened by two sequential validation stages. *Stage 1 — Statistical* (V1–V6) addresses class distribution and balancing (V1–V2), temporal splitting (V3), combinatorial purged cross-validation with PBO (V4), parameter-space variance (V5), and permutation testing (V6). *Stage 2 — Economic* (V7–V12) addresses net-of-cost performance (V7), cost sensitivity (V8), comparison against active baselines (V9), regime-conditional evaluation (V10), trade frequency (V11), and code reproducibility (V12). A strategy passing all 12 items is deemed VALID-compliant; failure at any item indicates a documented validation gap whose empirical prevalence is quantified in Section 4. Item rationales and corresponding failure modes are listed in Table 1.

After class balancing, long percentages drop to 32–52%, and balanced AUC converges to ≈ 0.50 , indicating no directional signal survives. Deep learning models (LSTM, Transformer) show less extreme bias (58–86% long) but produce AUC values similarly near random. The Transformer variant at the daily timeframe achieves the highest SR in our sample (+1.98)—but with $\text{PBO} = 1.0$, confirming overfitting (Section 4.5).

We extend this analysis to 18 asset-timeframe-model combinations (3 assets $\times$ 2 timeframes $\times$ 3 tree families). The

pattern is universal: mean unbalanced long percentages are 85% (BTC), 83% (ETH), and 80% (SOL). After balancing, mean AUC drops to 0.484 (BTC), 0.563 (ETH), and 0.542 (SOL)—all near random. Mean net SR values are -0.174 (BTC), $+0.023$ (ETH), and $+0.441$ (SOL), with all 1h models receiving $\text{PBO} = 1.000$. SOL’s positive mean SR is driven entirely by daily-timeframe variants that, as shown in Section 4.5, do not survive multiple testing correction.

Table 2: VALID versus existing ML reporting standards.

Standard	Domain	Items	Cost	Balance
TRIPOD+AI	Clinical ML	27	N/A	No
REFORMS	General ML	32	No	Partial
NeurIPS Checklist	ML research	15	No	No
VALID	Financial ML	12	Yes	Yes

Table 3: Literature audit results ($n = 74$ empirical papers). 95% Wilson confidence intervals.

Dimension	Failing	Rate	VALID
Costs omitted	40/74	54%	V7
No class balance	54/74	73%	V1, V2
Random split	5/74	7%	V3
CPCV used	0/74	0%	V4
BnH-only baseline	26/74	35%	V9
No net performance	40/74	54%	V7, V8
No code available	63/74	85%	V12

Table 4: Bull bias: directional prediction without and with class balancing (1h timeframe, 18 bp costs).

Asset (Model)	Unbal. L%	Bal. L%	AUC	Net SR
BTC (CatBoost)	97.2	42.3	0.50	-0.88
ETH (CatBoost)	90.5	45.2	0.49	-1.28
SOL (CatBoost)	97.0	32.2	0.51	-0.21
BTC (LSTM)	57.7	52.3	0.52	-1.15
BTC (Transformer) [‡]	—	—	0.64	+1.98

[‡]Daily TF, unbalanced only; no separate class-balancing experiment.

4.3 Cost Illusion and Regime Conditioning

Net performance degrades monotonically with trading frequency. This phenomenon is not unique to cryptocurrency: Novy-Marx and Velikov [22] documented similar cost erosion in equity anomalies, and Chen and Velikov [6] showed that after proper cost adjustment, the expected returns of most published anomalies converge to zero. Table 5 reports net Sharpe ratios at 18 bp for the model variants of the corpus, grouped by trading frequency. The pattern is stark: the median 15-minute variant returns $SR = -1.80$ and all nine lose money after costs, whereas the median daily variant returns $+0.54$. Only daily-frequency variants exceed the DM benchmark at all (best 1.98); the best hourly variant reaches 0.32. In the one run where gross and net Sharpe are directly comparable, transaction costs consume 82% of the gross Sharpe ratio ($0.750 \rightarrow 0.135$) at 101 trades per year.

Regime-conditional analysis using a two-state Hidden Markov Model (HMM) on 21-day rolling volatility reveals that the DM benchmark earns $SR = 1.40$ in stressed regimes (35% of evaluation days) versus $SR = 0.69$ in calm periods, while buy-and-hold achieves $SR = 0.79$ and 0.51 respectively. The benchmark’s alpha concentrates during market stress—precisely when practitioners most need reliable signals. This addresses concerns about bear-market bias: the benchmark outperforms

Table 5: Net Sharpe ratio (18 bp round-trip) of model variants in the 340-variant corpus, by trading frequency. Benchmark (DM): 0.917.

Frequency	n	Median	Best	Share < 0
15-minute	9	-1.80	-1.33	100%
1-hour	133	-1.32	+0.32	96%
4-hour	9	+0.28	+0.57	44%
Daily	77	+0.54	+1.98	22%

Model variants only; cost-sweep and rule-baseline rows of the corpus are excluded. Rule baselines: Section 4, Experimental Setup.

buy-and-hold in *both* regimes, with a spread of $+0.18$ in calm markets widening to $+0.61$ during stress.

4.4 Statistical-Economic Disconnect

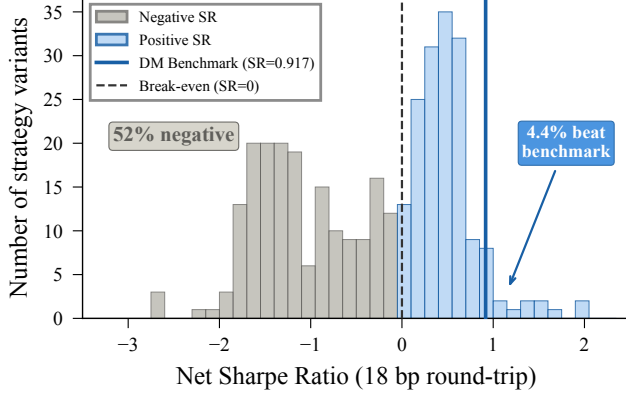
The order-flow 1h CatBoost run illustrates why Stage 1 requires more than significance testing. The model passes the permutation test decisively ($p = 0.000$; AUC 0.570 vs. a shuffled 95th percentile of 0.516): genuine signal exists. Its net SR of $+0.135$ against a benchmark of 0.917 already signals economic irrelevance — but a reviewer applying only significance tests would not see this coming. CPCV does: the same run’s PBO is 0.267, failing V4’s threshold. Selection on in-sample performance is not merely uninformative here but actively misleading: across CPCV configurations, the Spearman correlation between IS and OOS Sharpe is $\rho = -0.83$ ($p = 0.005$, $n = 9$ configurations), corroborated by the walk-forward analysis ($\rho = -0.73$, $n = 26$; Appendix G). The corpus-level version of this argument — nine Bonferroni survivors, all with PBO = 1.0, none surviving the Deflated Sharpe Ratio — follows in the next section.

Parameter-space variance (V5). For the 1-hour balanced pipeline (the base configuration without order-flow features), the mean within-fold variance of in-sample Sharpe across model configurations is $\text{Var}(SR_{IS}) = 0.012$, against a Monte Carlo null 95th percentile of 0.328. Here the two flatness criteria disagree (absolute threshold 0.01 vs. MC null 95th percentile 0.328), illustrating why V5 requires the null to be pre-specified. This provides empirical evidence consistent with Witzany’s [25] analysis of PBO under similar strategy returns.

This finding has a direct practical implication: **PBO alone cannot distinguish between a robust signal and a search space in which all configurations look alike.** V5 is a two-sided diagnostic. In-sample Sharpe variance far below the Monte Carlo null indicates a flat optimization landscape in which the search cannot distinguish configurations, while variance far above it signals parameter sensitivity and likely out-of-sample degradation. Neither extreme is desirable; V5 flags departure from the empirical null in either direction. An analytically derived acceptance threshold remains open.

Table 6: Ablation: false positive rates by VALID item removal.

Configuration	Removed	FPR	95% CI
Full VALID	None	0.0%	[0, 1.9%]
No CPCV/PBO	−V4	27.0%	[21, 34%]
No balancing	−V2	0.0%	[0, 5.8%]
No cost adj.	−V7	0.0%	[0, 1.9%]
No permutation	−V6	0.0%	[0, 1.9%]

**Figure 2: Distribution of net Sharpe ratios across 340 strategy variants (18 bp round-trip costs). Gray bars: negative SR (52%). Blue line: DM benchmark (SR = 0.917). Only 4.4% of variants exceed the benchmark, 10 of them model variants; after DSR correction, zero survive (Section 4.5).**

4.5 Ablation, Monte Carlo, and Multiple Testing

Ablation. Table 6 isolates the impact of individual VALID items by systematically removing each from the validation pipeline.

Removing V4 (CPCV/PBO) increases the false positive rate from 0% to 27%—the single most impactful validation component. No other item removal produces a measurable FPR increase, confirming V4’s critical role.

Monte Carlo. Table 7 reports false positive rates across four asset-timeframe settings using 200 synthetic null price series each (methodology in Appendix D).

AUC-based evaluation alone produces 27–30% false positives on signal-free data. CPCV with PBO eliminates all false positives across every setting—demonstrating that V4 is the single most effective component for avoiding type I errors in this context.

Multiple Testing Correction. Figure 2 shows the SR distribution across all 340 variants. Before correction, 15 variants (4.4%) exceed the simple momentum benchmark (SR = 0.917),

Table 7: False positive rates on null data ($n = 200$ per setting).

Criterion [§]	BTC 1h	ETH 1h	SOL 1h	BTC 1d
AUC > 0.55	27.0%	30.0%	26.5%	27.0%
Permutation*	22.0%	19.0%	18.0%	20.0%
PBO < 0.20	0.0%	0.0%	0.0%	0.0%
Net SR > 0	49.5%	52.0%	52.5%	49.5%
Full VALID	0.0%	0.0%	0.0%	0.0%

*BTC 1h permutation based on $n = 100$; all others $n = 200$.

[§]Column labels follow the original run names. All four settings were generated from daily-bar bootstrap series (838–919 events per series); the label mismatch is recorded in the repository’s REPRODUCE.md. The conclusion is unaffected: every setting yields 0% under CPCV with PBO.

Table 8: Strategies surviving Bonferroni correction ($N = 340$, $\alpha = 0.05$). All exhibit PBO = 1.0.

Strategy	SR	t	p_{Bonf}	AUC	Asset
Transformer ETH 1d	1.98	21.7	<0.001	0.64	ETH
RF SOL 1d (ext.)	1.94	20.5	<0.001	0.59	SOL
LightGBM ETH 1d	1.64	16.5	<0.001	0.64	ETH
CatBoost DXY 1d	1.57	17.4	<0.001	0.47	BTC
BiLSTM ETH 1d	1.46	13.0	<0.001	0.58	ETH
RF ETH 1d (ext.)	1.40	11.6	<0.001	0.63	ETH
CatBoost Gold 1d	1.37	12.6	<0.001	0.48	BTC
MLP SOL 1d	1.21	7.1	<0.001	0.52	SOL
TCN ETH 1d	1.08	4.1	0.008	0.57	ETH

p_{Bonf} : Bonferroni-adjusted ($p_{\text{raw}} \times 340$).

of which 10 are model variants; the remaining five are benchmark entries of the cost-level and baseline blocks. We apply five correction methods to assess robustness:

- **Bonferroni** ($N = 340$, $\alpha = 0.05$): 9 survive.
- **Holm-Bonferroni** (step-down): 9 survive.
- **Benjamini-Hochberg FDR**: 10 survive.
- **Harvey et al.** $t > 3.0$ [13]: 10 survive.
- **Deflated Sharpe Ratio** [5]: 0 survive.

Table 8 lists the nine Bonferroni survivors. All operate at the daily timeframe with t -statistics ranging from 4.07 to 21.70 (Lo [18] approximate adjustment). Cross-referencing with CPCV reveals the critical finding: *every single survivor exhibits PBO = 1.0*—complete backtest overfitting—and AUC values near random (0.47–0.64).

The Deflated Sharpe Ratio, which accounts for the expected maximum SR from N independent trials ($\mathbb{E}[\max] = 2.93$), independently rejects all 340 variants: the best observed SR (1.98) falls below the noise threshold.

This result demonstrates VALID’s central argument: **standard multiple testing correction is necessary but insufficient**. Classical corrections (Bonferroni, Holm, BH) control familywise or false-discovery error under independence assumptions, but they cannot detect backtest overfitting—strategies that are statistically “significant” by t -test yet demonstrably overfit by CPCV. The fact that a strategy can simultaneously satisfy $t > 21.0$ (extreme statistical significance) and $\text{PBO} = 1.0$ (complete overfitting) demonstrates

precisely why VALID Item V4 is indispensable: it catches what even the Harvey et al. $t > 3.0$ threshold misses.

5 Discussion

5.1 Scope Limitation: LLM-Based Trading Agents

All twelve VALID items are designed for structured, tabular prediction pipelines: triple-barrier labeling, combinatorial purging, and embargoing presuppose a feature-matrix workflow. Recent LLM-based trading agents fall outside this scope, since auditing non-deterministic generation, prompt sensitivity, semantic drift, and multi-agent feedback contamination requires programmatic criteria that do not yet exist. Extending VALID with quantifiable LLM-specific items is an open problem for future work; the present checklist should not be read as an audit instrument for LLM agents.

5.2 Limitations

Our analysis is subject to several limitations that we acknowledge explicitly to facilitate future extension.

First, the literature audit was conducted by a single coder without independent inter-rater reliability assessment. Unlike TRIPOD, which used a formal Delphi consensus process involving dozens of experts, VALID’s 12 items are derived from empirically quantified failure modes (Sections 4.2–4.5) rather than expert consensus. We provide the complete coding sheet and a detailed 10-paper subset in the appendix to enable independent verification; a formal Delphi validation study is planned as future work.

Second, our empirical validation covers three cryptocurrency assets (BTC, ETH, SOL). While the three failure modes—directional prediction bias, statistical-economic disconnect, and cost illusion—are not crypto-specific and arise in any trending, cost-sensitive asset class, extending to equities and foreign exchange would strengthen generalizability claims. SOL data begins in 2020, limiting historical depth; BTC serves as the primary asset, with ETH and SOL providing robustness checks.

Third, all 340 strategy variants use a single controlled pipeline design, analogous to ablation studies in deep learning research. This enables component-level attribution (e.g., Table 6’s isolation of V4 as the critical item) but may not capture failure modes specific to entirely different codebases or execution environments. Because variants share infrastructure, they are not independent; our Bonferroni correction is therefore conservative, and the true number of independent discoveries is likely fewer than nine—strengthening rather than weakening our conclusion that all survivors exhibit $PBO = 1.0$.

Fourth, our economic gates rely on a fixed, linear 18 bp round-trip cost proxy (Appendix F), which is a lower bound on realized frictions: it ignores volume-dependent market impact, order-book depth, and venue-specific slippage, which differ materially between BTC and lower-liquidity assets. Because all corpus variants are evaluated at a single 18 bp level,

V8 is only partially satisfied for our own pipeline (Appendices A, F); capacity constraints are explicitly out of scope, and integrating market-impact models into V7–V8 is left to future work.

In addition, VALID compresses continuous statistical evidence into binary items. Numeric cutoffs such as the 100-iteration minimum in V6 are reporting floors, not hypothesis tests: a study reporting 99 iterations with tight p -value bounds provides stronger evidence than a bare 100-iteration pass. We therefore recommend reporting the underlying statistics (p -values and confidence intervals) alongside each binary outcome, and treating the aggregate score as a screening device rather than a verdict.

Finally, our Monte Carlo analysis uses i.i.d. bootstrap resampling, which destroys temporal dependence. While this is by design (to create signal-free null data), block bootstrap alternatives would provide complementary evidence.

6 Conclusion

We propose VALID which, to our knowledge, is the first checklist-style protocol for financial ML that pairs binary statistical and economic gates in a fixed reporting order; it complements, rather than supersedes, prior backtesting guidance. Applying it to 340 strategy variants reveals that 52% produce negative net returns, standard validation (AUC alone) yields 27% false positive rates, and even strategies surviving multiple testing correction exhibit $PBO = 1.0$ —complete backtest overfitting. CPCV with PBO (Item V4) is the single most effective component for eliminating false positives in every tested setting. In a field where positive results are incentivized and negative results are invisible, standardized validation is the first step toward trustworthy financial ML research. We recommend that reviewers and program committees adopt VALID as a minimum reporting requirement for financial ML submissions, analogous to TRIPOD’s role in clinical prediction research.

Acknowledgments

Code and data are available at <https://github.com/orcajae/valid-framework>.

References

- [1] Gary Antonacci. 2014. *Dual Momentum Investing*. McGraw-Hill.
- [2] Hamid R. Arian, Daniel Norouzi Mobarekeh, and Luis Seco. 2024. Backtest overfitting in the machine learning era: A comparison of out-of-sample testing methods in a synthetic controlled environment. *Knowledge-Based Systems* 305 (2024), 112477.
- [3] Robert D. Arnott, Campbell R. Harvey, and Harry Markowitz. 2019. A backtesting protocol in the era of machine learning. *The Journal of Financial Data Science* 1, 1 (2019), 64–74. doi:10.3905/jfds.2019.1.064
- [4] David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. 2017. The probability of backtest overfitting. *Journal of Computational Finance* 20, 4 (2017), 39–69.
- [5] David H. Bailey and Marcos López de Prado. 2014. The deflated Sharpe ratio. *Journal of Portfolio Management* 40, 5 (2014), 94–107.
- [6] Andrew Y. Chen and Mihail Velikov. 2023. Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis* 58, 3 (2023), 968–1004.

- [7] Gary S. Collins, Karel G. M. Moons, Paula Dhiman, et al. 2024. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 385 (2024), e078378. doi:10.1136/bmj-2023-078378
- [8] Gary S. Collins, Johannes B. Reitsma, Douglas G. Altman, and Karel G. M. Moons. 2015. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD). *Annals of Internal Medicine* 162, 1 (2015), 55–63.
- [9] Jack Gallifant, Majid Afshar, Saleem Ameen, et al. 2025. The TRIPOD-LLM Reporting Guideline for Studies Using Large Language Models. *Nature Medicine* 31 (2025), 60–69. doi:10.1038/s41591-024-03425-5
- [10] Timnit Gebre, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (2021), 86–92. doi:10.1145/3458723
- [11] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. 2022. Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 35.
- [12] Peter Reinhard Hansen. 2005. A test for superior predictive ability. *Journal of Business & Economic Statistics* 23, 4 (2005), 365–380.
- [13] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the cross-section of expected returns. *Review of Financial Studies* 29, 1 (2016), 5–68.
- [14] John P. A. Ioannidis. 2005. Why most published research findings are false. *PLoS Medicine* 2, 8 (2005), e124.
- [15] Sayash Kapoor, Emily Cantrell, Kenny Peng, et al. 2024. REFORMS: Consensus-based recommendations for machine-learning-based science. *Science Advances* 10 (2024), eadk3452. doi:10.1126/sciadv.adk3452
- [16] Sayash Kapoor and Arvind Narayanan. 2023. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* 4, 9 (2023), 100804.
- [17] Florian Karl, Tobias Scherr, Theresa Pielok, Martin Binder, Florian Pfisterer, Stefan Coors, and Bernd Bischl. 2024. Position: Embracing Negative Results in Machine Learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Vol. 235. PMLR.
- [18] Andrew W. Lo. 2002. The statistics of Sharpe ratios. *Financial Analysts Journal* 58, 4 (2002), 36–52.
- [19] Marcos López de Prado. 2018. *Advances in Financial Machine Learning*. Wiley.
- [20] R. David McLean and Jeffrey Pontiff. 2016. Does academic research destroy stock return predictability? *Journal of Finance* 71, 1 (2016), 5–32.
- [21] Margaret Mitchell, Simone Wu, Andrew Zaldivar, et al. 2019. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. 220–229. doi:10.1145/3287560.3287596
- [22] Robert Novy-Marx and Mihail Velikov. 2016. A taxonomy of anomalies and their trading costs. *Review of Financial Studies* 29, 1 (2016), 104–147.
- [23] Halbert White. 2000. A reality check for data snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [24] Thomas V. Wiecki, Andrew Campbell, Justin Lent, and Jessica Staught. 2016. All That Glitters Is Not Gold: Comparing Backtest and Out-of-Sample Performance on a Large Cohort of Trading Algorithms. *The Journal of Investing* 25, 3 (2016), 69–80. doi:10.3905/joi.2016.25.3.069
- [25] Jiri Witzany. 2021. A Bayesian approach to measurement of backtest overfitting. *Risks* 9, 1 (2021), 18.
- [26] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. 2023. Are Transformers Effective for Time Series Forecasting?. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 11121–11128.

A VALID Self-Assessment

We apply VALID to our own study (Table 9).

All twelve items addressed: 9 fully satisfied, 3 partial (V5, V8, V10). V5 is context-dependent: the null distribution is derived rather than pre-specified, and the absolute and null-relative criteria disagree for our own pipeline (Section 4.4).

Table 9: VALID self-assessment of the present study.

#	Item	Status	Evidence
V1	Class distribution	✓	Table 4, Sec. 4.2
V2	Balanced testing	✓	Table 4, Sec. 4.2
V3	Temporal split	✓	CPCV with purge
V4	CPCV + PBO	✓	Tables 6–7, Sec. 4.5
V5	Var(SR _{IS})	~	Sec. 4.4 (criteria disagree)
V6	Permutation	✓	$n \geq 100$
V7	Net costs	✓	18 bp, Tables 4–5
V8	Cost sensitivity	~	Single 18 bp level; Sec. 4.3
V9	Baselines	✓	Sec. 4, setup (5 baselines)
V10	Regime eval.	~	Sec. 4.3 (3 assets)
V11	Trade frequency	✓	Sec. 4, setup; Sec. 4.3
V12	Code available	✓	Public GitHub repository

V8 is partial: all corpus variants are evaluated at a single 18 bp cost level, so we report a frequency ladder rather than a cost sweep of our own strategies. V10 is limited to three cryptocurrency assets; extension to equities and FX is planned. We note that this self-assessment was revised downward after auditing our own submission against VALID.

B Coding Subset

To enable independent verification of the literature audit (Section 4.1), we present the detailed coding for 10 randomly selected papers (seed = 42) across all seven dimensions. P = Pass, F = Fail.

Table 10: Coding subset: 10 randomly selected papers.

#	Authors	Year	D1 Cost	D2 Bal	D3 Split	D4 Val	D5 Base	D6 Net
13	Akyildirim+	2021	F	F	F	F	F	F
20	Borrageiro+	2022	P	F	P	F	F	P
35	Aboussalah+	2020	P	F	P	F	P	P
38	Ortu+	2022	F	F	P	F	P	F
41	Liew+	2022	F	F	P	F	P	F
44	Slepaczuk+	2023	P	F	P	F	F	P
45	Bu+	2023	F	F	P	F	P	F
51	Kose	2025	F	F	P	F	P	F
52	Sadorsky	2023	F	F	P	F	P	F
54	Sattarov+	2023	F	F	P	F	P	F
Pass rate			30%	0%	90%	0%	70%	30%

D7 (Code availability) is omitted for space; 1 of 10 papers (Aboussalah+) provides code. The complete 80-paper coding sheet is available in the public repository.

C Strategy Universe Detail

Table 11 presents the composition of the 340-variant strategy universe. All variants share a common pipeline (CUSUM event sampling, Triple Barrier labeling, temporal splitting with purge/embargo) and differ only in the component under investigation.

Components (a)–(c) test whether additional feature engineering improves predictive performance beyond the base

Table 11: Composition of the 340-variant strategy universe.

ID	Component Description	n
(a)	SHAP-derived rule combinations	64
(b)	Order flow features (individual + multi-asset)	55
(c)	External/on-chain/macro feature sets	45
(d)	ML pipeline: 3 assets $\times$ 4 TF $\times$ 3 trees	36
(e)	Triple Barrier parameter sensitivity	36
(f)	Cross-section momentum configurations	24
(g)	Cost sensitivity & traditional baselines	24
(h)	Deep learning: 3 assets $\times$ 2 TF $\times$ 5 arch.	30
(i)	VALID ablation configurations	12
(j)	Volatility prediction overlays	8
(k)	Bull bias analysis (tree + DL)	6
Total unique configurations		340

pipeline. Components (d)–(e) vary the core pipeline across assets, timeframes, model families, and labeling parameters. Components (f)–(g) provide cross-section and cost-sensitivity analyses. Component (h) extends to five deep learning architectures (Transformer, LSTM, BiLSTM, TCN, MLP with 2 and 3 hidden layers). Components (i)–(k) support the VALID framework evaluation directly.

D Monte Carlo Methodology

D.1 Synthetic Data Generation

We generate $n = 200$ synthetic BTC price series by bootstrapping daily log-returns (2017–2026, 3,132 observations per series). The bootstrap preserves the marginal distribution of returns (mean, variance, skewness, kurtosis) while destroying temporal dependence, ensuring that any predictive signal detected by the ML pipeline is spurious by construction. We use simple i.i.d. resampling rather than block or stationary bootstrap, as our goal is to eliminate all temporal structure. The choice of $n = 200$ yields Wilson CI half-widths of $\pm 6\%$ at $\text{FPR} = 27\%$, providing sufficient precision to distinguish informative from uninformative validation criteria.

D.2 Pipeline Application

Each synthetic series passes through the identical pipeline used for real data:

- (1) Compute 17 price-derived technical features (returns, volatility, momentum indicators).
- (2) Generate Triple Barrier labels with CUSUM filtering.
- (3) Train CatBoost with class-balanced sample weights.
- (4) Evaluate via CPCV ($N = 6$, $k = 2$, yielding 15 train-test combinations).
- (5) Record AUC, PBO, $\text{Var}(\text{SR}_{\text{IS}})$, permutation p -value, and net SR (18 bp round-trip cost).

The pipeline is deterministic given the input data; no hyperparameter tuning is performed within the Monte Carlo loop. This ensures that any false positive arises from the validation criterion, not from adaptive model selection.

E Mathematical Definitions

E.1 Triple Barrier Labeling

Following López de Prado [19], each observation is assigned a label $y_i \in \{-1, 0, +1\}$ based on the first barrier touched by the price path after event time t_i . Let p_t denote the price at time t . Three barriers are defined:

$$\text{Upper: } p_{t_i+\tau} \geq p_{t_i}(1 + \theta_u) \quad (1)$$

$$\text{Lower: } p_{t_i+\tau} \leq p_{t_i}(1 - \theta_l) \quad (2)$$

$$\text{Vertical: } \tau = T_{\max} \quad (3)$$

where $\theta_u, \theta_l > 0$ are profit-taking and stop-loss thresholds scaled to local volatility ($\theta_u = 2.0\hat{\sigma}$, $\theta_l = 2.5\hat{\sigma}$ in our daily configuration), and $T_{\max}$ is the maximum holding period. Events are sampled via CUSUM filtering on cumulative log-returns with threshold h .

E.2 Probability of Backtest Overfitting (PBO)

$$\text{PBO} = \frac{1}{\binom{N}{k}} \sum_{j=1}^{\binom{N}{k}} \mathbf{1}[\text{logit}(\bar{\omega}^{(j)}) \leq 0] \quad (4)$$

where $\text{logit}(x) = \ln(x/(1-x))$ and $\bar{\omega}^{(j)}$ is the relative rank of the in-sample-best configuration in the out-of-sample ranking for combination j . $\text{PBO} \approx 0$ indicates that in-sample-best configurations tend to rank in the top half out-of-sample; $\text{PBO} \approx 1$ indicates severe overfitting [4].

E.3 Parameter-Space Variance

$$\text{Var}(\text{SR}_{\text{IS}}) = \frac{1}{M-1} \sum_{m=1}^M \left(\text{SR}_{\text{IS}}^{(m)} - \overline{\text{SR}_{\text{IS}}} \right)^2 \quad (5)$$

Low $\text{Var}(\text{SR}_{\text{IS}})$ relative to a null distribution indicates that all configurations perform similarly, rendering $\text{PBO} = 0$ uninformative rather than reassuring [25].

E.4 Sharpe Ratio to t -Statistic Conversion

Following Lo [18], the t -statistic for testing $H_0 : \text{SR} = 0$ is:

$$t = \frac{\text{SR} \cdot \sqrt{T}}{\sqrt{1 + \frac{1}{2} \text{SR}^2}} \quad (6)$$

where T is the number of return observations. This is the i.i.d. approximation; the full adjustment includes skewness and kurtosis terms. We use this formula for the multiple testing correction in Section 4.5, applied to excess SR over the benchmark.

F Cost Model Detail

Our 18 bp fixed round-trip cost is conservative for hourly and daily trading frequencies. It comprises:

- **Exchange fee:** Binance VIP-0 taker fee of 10 bp (0.10%) per side = 20 bp round-trip before fee token discount. With BNB discount: ≈ 15 bp round-trip.

- **Slippage:** Estimated 5–10 bp per round-trip for market orders of moderate size (\$1K–\$10K) on BTC/USDT during liquid hours.
- **Net:** 18 bp is a rounded conservative estimate that accounts for both components.

For sub-hourly frequencies, volume-dependent slippage dynamics become significant and would require Almgren-Chriss optimal execution models, further eroding returns and strengthening rather than weakening our conclusions. All corpus variants are evaluated at a single 18 bp level, so we make no robustness claim across a cost range for our own strategies; V8 is scored as only partially satisfied in Appendix A. The frequency ladder of Table 5 and the 82% gross-to-net erosion of Section 4.3 both indicate that higher cost assumptions would strengthen, not weaken, the conclusions drawn here.

G Validation Method Comparison

AUC-based validation produces 27–30% false positives on null data; CPCV with PBO < 0.20 produces 0% (Table 7). An expanding-window walk-forward (3-year train, 6-month test, 3-month slide) produces 32 windows. After removing 6 windows with extreme OOS SR (>10, caused by circuit-breaker edge cases at period boundaries), the walk-forward IS-OOS Spearman correlation is $\rho = -0.73$ ($p < 0.001$, $n = 26$): higher in-sample performance predicts lower out-of-sample performance. Across the nine CPCV model configurations, the IS-OOS Spearman correlation is $\rho = -0.83$ ($p = 0.005$), consistent with the finding of Wiecki et al. [24] of $R^2 < 0.025$ across 888 Quantopian strategies. Both statistics are configuration-level; we do not report a fold-level correlation.

Computational cost. The full 340-variant pipeline runs in approximately 8 hours on an Apple M4 with 24 GB RAM. The 200-iteration Monte Carlo analysis adds ~12 hours per asset-timeframe setting. Multiple testing correction completes in under 5 minutes.